SAR-TSCC: A Novel Approach for Long Time Series SAR Image Change Detection and Pattern Analysis

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***Abstract*— Change detection has played an increasingly important role in multitemporal remote sensing applications recently. Long time series analysis is providing new information of land cover changes and improving the quality and accuracy of the change information being derived from remote sensing. The purpose of this study is to dig for more change temporal information and change pattern information from synthetic aperture radar (SAR) image time series (ITS), which is of great significance for monitoring urban area changes, conducting land use surveys, and renovating illegal constructions. In the study, a novel unified framework for long time series SAR image change detection and change pattern analysis (SAR-TSCC) was proposed for land cover change mapping. To obtain the most notable change time rapidly, a fast SAR ITS change point search method based on pruned exact linear time (SAR-PELT) algorithm was adopted. Meanwhile, the deep time series classification network, named SAR time series transformer (SAR-TST), was implemented to recognize the change patterns, which is based on time series transformer (TST) architecture. Considering the lack of real training data, a novel synthetic data generation method is developed. The combination of the synthetic and real data enhanced the generalization of the classifiers. The proposed framework was used for monitoring a large urbanization area in the northwest of Hong Kong, China. The Cosmo Skymed (CSK) time series data acquired from 2013 to 2020 were exploited for land cover change analysis. Experiment results showed that our approach achieved the state-of-the-art performance, as the time accuracy reached 86% and the classification accuracy on the four main change patterns (impulse, step, cycle, and complex) is over 99%. In particular, the proposed SAR-TST model showed remarkable advantages in the presence of insufficient real data.**

***Index Terms*— Change detection, deep learning (DL), synthetic aperture radar (SAR) images, time series.**

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I. Introduction

CHANGE detection is a process of using multitemporal remote sensing images covering the same regions at different dates to identify the changes. As a significant application of remote sensing, change detection technology has been widely implemented in urban area monitoring [1], [2], [3], agriculture survey [4], [5], [6], forest monitoring [7], [8], [9], and disaster assessment [10], [11], [12], [13], [14], [15], [16]. In particular, with the rapid expansion of urban areas, land cover and land use (LCLU) changes are needed to be monitored effectively. Cracking down on acts of illegal construction and demolition has become an important part of urban management and law enforcement. Monitoring the location, extent, and frequency of LCLU transitions and recognizing the temporal patterns of change through automated interpretation based on high-resolution remote sensing images are valuable to aid in learning the causes and drivers of change. Compared with optical sensors, synthetic aperture radar (SAR) works under nearly all weather conditions and at any time, and has been indeed widely exploited as an important data source for Earth observation [17]. Benefiting from the rapid development of SAR satellite remote sensing technology, a large amount of multitemporal images is easily available, which drives change detection transforming from bitemporal analysis to density long time series analysis. Using SAR image time series (ITS) to rapidly determine the precise change time and identify the patterns of the change has been the focus of major research [18]. It is of critical importance to provide urban managers with timely and precise LCLU change dynamics and decision-making suggestions [19].

In recent years, the rapid development of time series analysis technology has driven the progress of SAR ITS change detection. At present, change detection methods based on time series analysis mainly focus on acquiring single change information, e.g., spatial information [20] and temporal information [21]. Although there are novel visualization

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models, such as REACTIV algorithm [22], simultaneously incorporating change frequency and change time, it is still lack the description of change patterns. Meanwhile, the present change pattern recognition methods, e.g., K-Matrix [23] and PMCMA [24], are mainly based on unsupervised clustering strategies, which are only applicable to very short time series. Also, deep learning (DL) methods have not yet shown their capabilities in this field up to now [25].

There are mainly the following problems in previous works for SAR ITS change detection.

1) The accuracy of change temporal information has yet to be improved. Especially for long time series SAR data, the existence of speckle noise makes it easy to generate abnormal points, which always causes high error detection.
2) The classification precision of the change patterns is still far from satisfactory. The existing unsupervised clustering methods are sensitive to parameters and generally not robust to the speckle noise, yielding poor classification performance on long time series and complex scenes [23], [24], [25].

For the problems above, there is a lack of an effective method to obtain the exact change time and recognize the change temporal patterns by exploiting the information provided by the long SAR ITS. Therefore, in this study, we proposed novel methods that tackle the issues above. The main contributions are summarized as follows.

1) A fast search strategy based on pruned exact linear time (PELT) change point detection (CPD) algorithm, called SAR-PELT, was developed to extract the most notable change point over the detected time series. Compared with traversal comparison methods, the proposed algorithm is speedy and efficient, as well as more robust to abnormal points.
2) To recognize the change patterns effectively, a deeply supervised pattern recognition model named SAR time series transformer (SAR-TST) is designed. The pattern recognition task is transformed into a time series classification (TSC) problem. To obtain sufficient pretraining samples, a synthetic data generation method is developed based on a time series composition theory. The classifier was first pretrained on synthetic data, and then, the model weights were fine-tuned using a small amount of labeled real-world data, which achieved high classification accuracy on the four main types of change patterns, i.e., impulse, step, cycle, and complex.

The Cosmo Skymed (CSK) time series data acquired from 2013 to 2020, covering the northwest of Hong Kong, were exploited to verify the practicality of the proposed framework. Such framework detected change time with an overall accuracy (OA) of 86% and classified change patterns with a high OA of 99%, which confirms the strong generalization and stability in monitoring large-scale complex urban areas. The primary novelties of the study are: 1) implementation of the CPD method for rapid LCLU mapping and 2) development of an automated interpretation method to identify the typical change temporal patterns.

The rest of this article is organized as follows. Section II briefly introduces the related works on SAR image change detection. Section III presents the proposed framework in detail. The extensive experimental results and detailed discussion are presented in Section IV. Finally, we draw the conclusion for this article in Section V.

II. Related Work

In the following, we will present a detailed overview of the recent change detection approaches based on SAR images, where these change detection methods can be further categorized into bitemporal- and multitemporal-based methods according to the number of input images to be analyzed.

A. Bitemporal-Based Change Detection Method

Bitemporal-based change detection methods usually detect space information from two-date images, including change location and extent. Recently, a series of bitemporal-based methods were developed from pixel level to semantic level. These strategies are involved in ratio operator [26], image fusion [27], [28], [29], feature transforming [30], [31], [32], similarity measurement [33], [34], and so on. DL-based methods, which have the powerful feature learning ability, also showed great potential for application [35], [36], [37]. Most of them are focused on extracting the changed regions from the background with higher accuracy rather than on what the change types are. The essence of the bitemporal analysis is to obtain the differences of the backscattering between the acquired images from two dates. However, it may cause a lot of false alarms and miss detection when the differences are slight. In addition, it is a difficult task to detect the small and dense changes, such as some changes in denseness residential areas. Bitemporal change analysis cannot insights on the nature of the transformations, such as change patterns, since it has the inherent defect on making full use of temporal information.

B. Multitemporal-Based Change Detection Method

Multitemporal-based time series analysis has proven to address the challenges outlined above, as it can obtain the extra temporal evolution information in the specific time interval. An effective method is pairwise traversal comparison. The kernel is to build a change detection matrix (CDM), and each pixel stack is assigned a CDM containing the binary change information over the time series. In [38], the CDM constructed by similarity cross tests is exploited for adaptive temporal filtering, analysis of change dynamics, and multitemporal change detection. Su et al. [18] proposed an SAR ITS change analysis method called NORCAMA. The authors first used a multitemporal filtering method to suppress the speckle noise, and a generalized likelihood ratio test was adopted to calculate the statistical similarity between every two dates. The binary test results are then used to construct the change criterion matrix (CCM). Finally, a normalized cut-based clustering method is implemented on CCM to identify four main change patterns. Lê et al. [39] also introduced CDM to monitor floods that happened in the central coast regions of Vietnam. The authors proposed a novel multiscale framework to rapidly obtain the change location and change patterns.

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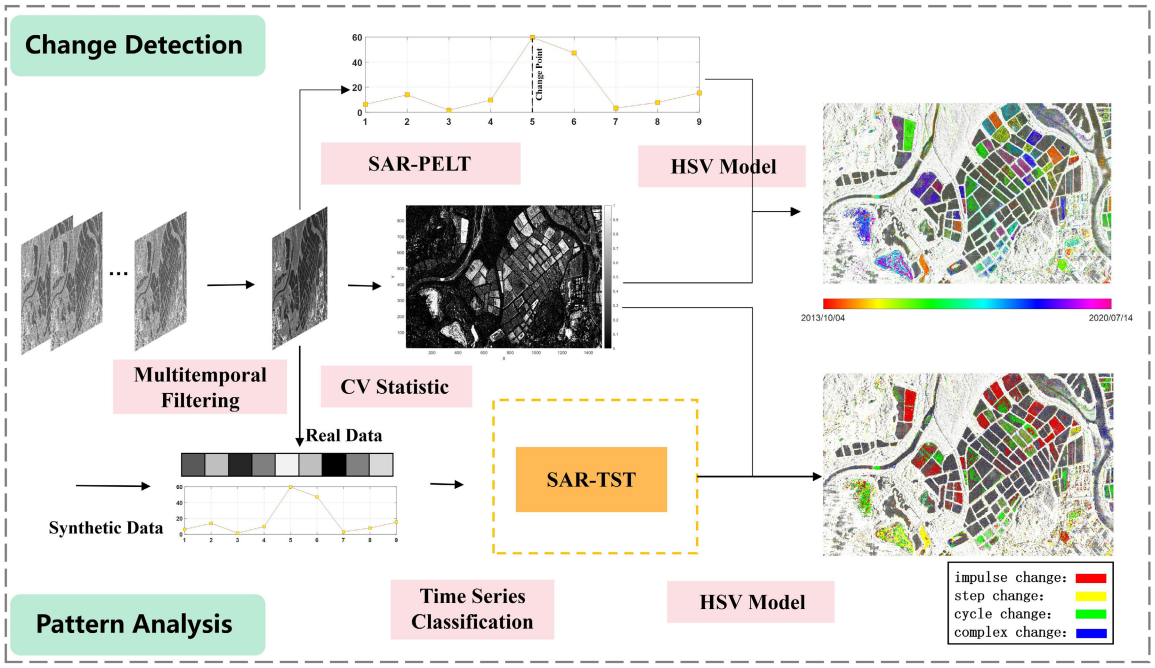


Fig. 1. Schematic of the proposed SAR ITS change detection and pattern analysis framework.

Compared with the methods mentioned above, statistical modeling on the whole time series is more direct. The MIMOSA method proposed by Quin et al. [20] is based on the computation of the theoretical joint probability density function between the geometric and quadratic means of SAR ITS. Automatic threshold according to a given false alarm rate is the only parameter to detect the changes. In [21], the temporal coefficient of variation computer vision (CV) for SAR ITS has been certified to be an effective change frequency indicator for detecting urban area dynamics. Jiang et al. [40] also used CV to monitor the built-up land changes. Besides, DL-based methods have made some progress in this field. Yan et al. [41] proposed a deep transfer learning model based on similarity measurement, called SDTL-TSACD, which has been implemented for monitoring land cover changes on the Loess Plateau. Saha et al. [42] transferred change detection into an anomaly detection problem and exploited a multilayer long short-term memory (LSTM) network to learn a representation of the time series. The model has been successfully applied to two small-scale Sentinel-1 datasets.

Through the above review of the recently developed SAR ITS change detection methods, the following conclusions can be drawn.

1) Change analysis over the temporal profiles is showing a vast research potential, as the temporal resolution of SAR ITS is continuously improved. However, the current time series analysis methods are too complex to fast apply to large-scale SAR images. It is necessary to develop a fast method that can extract change time information to map the LCLU changes.
2) DL-based models are showing great power in time series analysis, which also injects fresh blood into SAR ITS analysis. The introduction of DL-based methods could lead to higher accuracy on change pattern classification. Nonetheless, the availability of adequate labeled data and complicated network designs are key factors in the success of DL-based approaches. However, there are inevitably suffering from the lack of real-world labeled samples in SAR ITS application since labeling large amounts of labeled data is time-consuming and requires expertise. Therefore, developing a powerful transfer model that utilizes the sufficient synthetic data is an effective way to solve the above problem.

III. Methodology

Let $I = \{I_t\}_{1 \le t \le N}$ be a set of SAR images acquired over the same geographical region at different times. Some regions in the scene might experienced diffierent types of change in this period. The purpose of the study is to find the LCLU change time quickly and recognize the change types effectively. The flowchart of the proposed SAR-TSCC framework is shown in Fig. 1. It mainly includes two parts: 1) the SAR-PELT method is used to extract the change time information and 2) the SAR-TST model is adopted to recognize the change temporal patterns. Both the results are visualized by the hue, saturation and value (HSV) color space model [21]. The following will give the specific introduction.

A. Data Preprocessing

In this study, we first applied a pure geometric algorithm for coregistration of the CSK data with the aid of the Shuttle Radar Topography Mission (SRTM) 3 arc-second digital elevation model (DEM) [43], [44]. The details of this method can be found in [45]. All the images used in this study were coregistered to precisely stack all pixels located on the same position. In addition, multitemporal filtering is also a necessary step during preprocessing. The abnormal points over the time series

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caused by the speckle noise could interfere with time series analysis. To suppress the interference from the speckles, the ratio-based multitemporal SAR image denoising (RABASAR) algorithm is introduced in this study. As a multitemporal denoising method, RABASAR uses a "superimage" to combine the spatial filtering and temporal averaging. Meanwhile, the denoiser called RuLoG is implemented in the ratio image formed between the current image and the "superimage." The filtered image is recovered by multiplying the denoised ratio with the superimage. Here, we briefly introduce the main steps of this algorithm, and the details can be found in [46]. The introduction of RABASAR eliminates the strong speckle noise, which can help with the exact implementation of the following downstream tasks.

B. Information of Change Frequency

Those areas with high-frequency changes, which shows high fluctuations over time series, are always the focus to monitor. Coefficient of variation, as a commonly used metric of relative dispersion, has been employed in measuring the extent of the temporal variability of SAR ITS. Given an SAR image stack containing N images, all of them have been preprocessed. For a pixel series at spatial location $\mathbf{s} = \{s_t\}_{1\le t\le N}$, simply denoted $s_{1:N}$, the standard expression of CV is

$$\mathrm{CV}_s = \frac{\hat{\sigma}_s}{\hat{\mu}_s} \tag{1}$$

where $\hat{\mu}$ and $\hat{\sigma}$ indicate the mean and standard deviation of the pixel series, respectively. It can also be expressed by the first two moments of any statistical distribution

$$\mathrm{CV}_s = \frac{\sqrt{m_2 - m_1^2}}{m_1} \tag{2}$$

where m_1 and m_2 are the first- and second-order moments, respectively. The amplitudes of the pixel series in a speckle area follow a Rayleigh–Nakagami law [20]:

$$\mathcal{RN}[\mu, L](u) = \frac{2}{\mu}\frac{\sqrt{L}}{\Gamma(L)}\left(\frac{\sqrt{L}u}{\mu}\right)^{2L-1} e^{-\left(\frac{\sqrt{L}}{\mu}u\right)^2} \tag{3}$$

where μ represents the shape parameter, L represents the equivalent number of looks (ENL), and Γ represents the Gamma function. Then, the first- and second-order moments can be both derived under the Rayleigh–Nakagami law. The expressions are

$$m_1 = \mu\frac{\Gamma(L+\frac{1}{2})}{\sqrt{L}\Gamma(L)} \tag{4}$$

$$m_2 = \mu^2. \tag{5}$$

With (4) and (5), (2) can be rewritten as

$$\mathrm{CV} = \frac{\sqrt{m_2 - m_1^2}}{m_1} = \sqrt{\frac{\Gamma(L)\Gamma(L+1)}{\Gamma(L+1/2)^2} - 1}. \tag{6}$$

For the temporal stable speckle area, in particular, when the speckles are fully developed, $L = 1$, and the result calculated from (6) is 0.5227. When it comes to the permanent scatterer, which is stable and almost unchanged, the CV is far lower than 0.5227 due to the higher SNR. On the contrary, CV will be very high in the area containing temporal changes, as $\hat{\sigma}$ in (1) has a sharp rise. CV is, therefore, to be an excellent indicator to measure the frequency of the changes. It can be considered that the higher the CV value, the more fiercely the land changes.

TABLE I
PROCESS OF SEARCHING CHANGE POINTS USING SAR-PELT

Algorithm 1 SAR-PELT algorithm for SAR ITS

Input: Pixel series $s_{1:N}$;Cost function $\mathcal{C}(\cdot)$; Penalty value β
Initialize: Criterion function:$F(0) = -\beta$;
Change point indexes:$cp(0) = NULL$;
Admissible indexes:$R_1 = \{0\}$;
for $\tau^* = 1, \cdots, N$ **do**
  $F(\tau^*) = min_{\tau\in R_{\tau^*}}[F(t) + \mathcal{C}s_{(t+1):\tau^*} + \beta]$
  $\tau^1 = arg\left\{min_{\tau\in R_{\tau^*}}[F(t) + \mathcal{C}s_{(t+1):\tau^*} + \beta]\right\}$
  $cp(\tau^*) = [cp(\tau^1), \tau^1]$
  $R_{\tau^*+1} = \left\{\left\{\tau \in R_{\tau^*} : F(\tau) + \mathcal{C}s_{(t+1):\tau^*} \le F(\tau^*)\right\} \cup \tau^*\right\}$
end
return $cp(N)$
Calculate the $\tau^{max} = max_{\tau^i\in cp(N)}[s(\tau^i) - s(\tau^i - 1)]$
Output: The change point index τ^{max}

C. Information of Change Time

The temporal statistical properties will correspondingly be changed when the land change occurs in SAR ITS. An effective way to get the change time information is using the CPD strategy. However, the number of change points in the time dimension is unknown. The PELT algorithm, as an adaptive CPD method, can recognize the significant change points fast without any prior information [47]. Based on the PELT algorithm, a novel rapid SAR ITS change time search method, which we call SAR-PELT, is developed in the study.

The flowchart of the SAR-PELT algorithm is shown in Table I. For the pixel series $s_{1:N}$, we hope to find the number of the change points m and their time indexes $\tau_{1:m} = (\tau_1, \tau_2, \ldots, \tau_m)$. Each index is an integer between 1 and $N-1$ inclusive. We define $\tau_0 = 0$ and $\tau_{m+1} = N$ and assume that the change points are ordered. The pixel series will be split into $m+1$ segments, where the ith segment denotes $s_{(\tau_{i-1}+1):\tau_i}$. Then, finding the multiple change points can be transformed into an optimization problem. The optimized object function is constructed as

$$\sum_{i=1}^{m+1}[\mathcal{C}\left(s_{(\tau_{i-1}+1):\tau_i}\right) + \beta f(m)] \tag{7}$$

where $\mathcal{C}(\cdot)$ is the cost function, $\beta f(m)$ is the added penalty to guard against overfitting, and β is a smoothing parameter to control the weight of the penalty. In this article, the cost function is based on the empirical standard derivation

$$\sum_{i=p}^{q}\Delta(s_i; \chi([s_p \cdots s_q])) = (q-p+1)\log\sum_{i=p}^{q}\sigma^2([s_p \cdots s_q]) \tag{8}$$

where σ^2 indicates the standard derivation of the subseries $s_{p:q}$. To simplify and accelerate the solving process, let $F(s)$

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denote the optimal value of the criterion function from (7). Then, it follows that:

$$\begin{aligned} F(s) &= \min \sum_{i=1}^{m+1} [\mathcal{C}\left(s_{(\tau_{i-1}+1):\tau_i}\right) + \beta] \\ &= \min_t \left\{ \min \sum_{i=1}^{m} [\mathcal{C}\left(s_{(\tau_{i-1}+1):\tau_i}\right) + \beta] + \mathcal{C}s_{(t+1):n} + \beta \right\} \\ &= \min_t \left\{ F(t) + \mathcal{C}s_{(t+1):n} + \beta \right\}. \end{aligned} \tag{9}$$

Formula (9) demonstrates that the introduction of a change point will reduce the cost function. Extracting the set of change points in the optimal segmentation is achieved by a simple recursion backward through the time series, which significantly improves the calculating efficiency.

Several change points can be obtained using the above rapid searching method. After that, the most significant change point can be extracted using the neighborhood difference comparison strategy

$$\Delta_\tau s(\tau_i) = s(\tau_i) - s(\tau_i - 1). \tag{10}$$

D. Information of Change Patterns

Identifying the change temporal patterns is helpful to understand the nature of LCLU transitions and master the dynamic change regularity. In this article, we divided the patterns into four main types: impulse, step, cycle, and complex. The impulse change is tending to be a type of transient and temporary e.g., moving vehicles and sudden floods. The step change is a persistent or a permanent type, such as new constructions. The cycle change is often seen as a seasonal or regular type, such as seasonal crop areas. The complex change is represented as a chaotic pattern.

From the perspective of DL, recognizing the typical change patterns from the SAR ITS can be seen as a TSC problem. However, the main challenge of this task is limited training samples, as it is difficult to obtain sufficient labeled data on SAR ITS [48]. Synthetic data have been increasingly used in DL applications to overcome the few-shot problem [49], [50], [51], [52]. SAR interferometry (InSAR) synthetic time series samples have been used for classifying the diverse deformation patterns [53]. Simulated data have also been applied to SAR target recognition [54], [55]. In this study, we obtained sufficient simulated samples from a novel synthetic data generation method to overcome the few-shot problem.

1) Synthetic Data of SAR ITS: The novel synthetic data generation method is based on time series composition theory. For a nonstationary random sequence $Y(t)$, we first assume it as an additive decomposition model. Then, it can be decomposed into three parts according to the seasonal-trend decomposition procedure based on loess (STL) algorithm [56], which can be expressed as

$$Y(t) = T(t) + S(t) + R(t) \tag{11}$$

where $T(t)$ indicates the trend component, $S(t)$ indicates the seasonal component, and $R(t)$ indicates the remainder component. Similarly, one multiplicative decomposition model can be transformed into an additive by processing with logarithm. Based on this decomposition method, SAR ITS can also be correspondingly decomposed into three parts: backscattering component $I(t)$, change component $C(t)$, and the speckle noise component $N(t)$. Then, (11) is updated as

$$Y(t) = I(t) + C(t) + N(t). \tag{12}$$

It should be noted that it is difficult to obtain $I(t)$, which is closely coupled with $N(t)$. Therefore, they can be amalgamated into a background component $B(t)$ and (12) can be rewritten as

$$Y(t) = B(t) + C(t). \tag{13}$$

For $B(t)$, it can be seen as the speckle area without any changes, and thus, it follows the Rayleigh–Nakagami law according to (3). For $C(t)$, it was considered a sudden change rather than a long-term trend. The impulse change pattern can be simply expressed as

$$C(t_i)_{\text{impulse}} = A \cdot \text{mean}(B(t_i)), \quad t_0 \le t_i \le t_1 \tag{14}$$

where A denotes the amplitude parameter and t_1–t_0 is the duration of the change. The cycle change can be obtained by setting multiple impulse periods. Similarly, the step change can be obtained by adjusting the interval of t_i

$$C(t_i)_{\text{step}} = A \cdot \text{mean}(B(t_i)), 0 \le t_i \le t_1 \text{ or } t_1 \le t_i \le t_{\max}. \tag{15}$$

As for the complex changes, the waveform of the time series is chaotic. It can be expressed by enlarging the background component $B(t)$

$$C(t_i)_{\text{complex}} = A \cdot B(t_i), \quad 0 \le t_i \le t_{\max}. \tag{16}$$

Based on the simplified synthetic expressions above, various change patterns can be synthesized with different change amplitudes and cycles. If parameter A is set to values smaller than 3, we found that the changes are easily masked by the speckle noise. Therefore, to highlight the changes and broaden the differences among the selected four change types, in this study, the values of A are randomly selected in the range between 3 and 5, according to a uniform distribution. Moreover, since the durations and positions of the changes are hard to be simulated, our model is simpler. Specifically, the change center points for impulse change are uniformly distributed along the time series. The durations of two-thirds of the samples are randomly selected between 3 and 10 and the others are randomly selected between 11 and 60. Both parameters are combined randomly, and those samples whose impulse boundaries exceed the time series length are discarded. These settings are also applied to cycle change samples. The extra parameter is the number of cycles, which is uniformly and randomly selected between 2 and 4. For the step change, the transition duration is selected between 3 and 60. Fig. 2 shows a few synthetic sequences. It is worth noticing that synthetic data cannot entirely replace real-world data, as they are obtained from an approximate model. In the experiments reported in Section IV, the differences are evaluated in detail.

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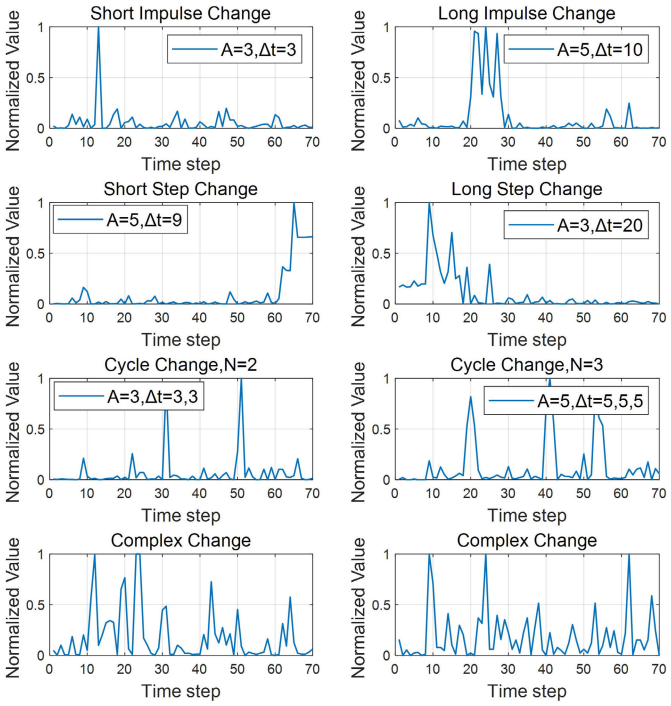


Fig. 2. Some cases of different change pattern synthetic samples. A denotes the amplitude parameter, and Δt is the duration of the change impulse.

2) SAR ITS Classification Model: In this work, we use the transformer model to conduct the SAR ITS change pattern classification task. Transformer, as a recently developed DL model, completely distinguishes from traditional convolutional neural network (CNN) and recurrent neural network (RNN) architectures, which extracts the features using the novel multihead attention module [57]. It has been monopolizing the state-of-the-art performance across all natural language processing (NLP) and CV tasks. Recently, it has reached or approached the state-of-the-art level in time series tasks, such as prediction and classification [58].

SAR ITS has strong temporal correlations. From the temporal perspective, the time series fluctuation of the backscatter contains intrinsic information about change patterns, which is showing significant regional differences. Although the DL-based methods have been applied to SAR ITS change detection [42], they have yet to be implemented in change pattern analysis up to the present. Inspired by the time series self-supervised representation learning, we tried to expand their applications into the change pattern recognition field. The training mode used in our proposed SAR-TST is similar to bidirectional encoder representations from transformers (BERT) [59]. First, the self-unsupervised pretraining strategy is conducted on massive unlabeled synthetic data to capture the generic backscatter-temporal features. The fine-tuning step is then applied to the initialized model using limited labeled real-world data captured from the study area. The overall framework is shown in Fig. 3. The combination of synthetic data and real data can learn the stable change temporal patterns and is more robust to the speckle noise.

The base network architecture of SAR-TST is shown in Fig. 4. The core of SAR-TST is the encoder module from the original transformer architecture, but the decoder part is not preserved. As a discrimination module, the independent encoder is suitable for classification tasks. Meanwhile, discarding the decoder part reduced half the model parameters, which could avoid the potential overfitting. As shown in Fig. 4, the bottom is the input module. Specifically, for an input training sample $\mathbf{s} = (s_1, s_2, \ldots, s_N)$, we first normalize it by subtracting the mean and dividing by the variance. Then, they are linearly projected onto a new vector space (input embedding)

$$\mathbf{u}_t = w\mathbf{s}_t + b \tag{17}$$

where $\mathbf{u}_t \in \mathbb{R}^d$, $t = 1, \ldots, N$ are the new feature vectors after conducting a d-dimensional projecting, which is similar to the word vectors in NLP, and $w \in \mathbb{R}^{d\times 1}$ and $b \in \mathbb{R}^{d\times 1}$ are both learnable parameters. Then, we add position encoding to $\mathbf{u}_t$, which is a necessary step to get the sequential features

$$\mathbf{u}' = \mathbf{u} + W_{\text{pos}} \tag{18}$$

where $\mathbf{u} = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_N] \in \mathbb{R}^{N\times d}$ and $W_{\text{pos}} \in \mathbb{R}^{N\times d}$ is the fully learnable position encoding rather than the original deterministic sinusoidal encoding, which can learn the sequential features more effectively.

After processing with input encoding, the input vectors contain the location information of each time point and they will become the queries, keys, and values in the self-attention module of the transformer encoder. The backbone of the SAR-TST includes N-layer transformer encoders, where every encoder takes the feature representations from the previous layer as its input and iteratively generates higher level feature representations to pass to the next layer. A single transformer encoder block consists of a multihead attention module and a fully connected feedforward sublayer. The residual connection and layer normalization are both utilized to improve the performance of the model. Finally, the fully connected output layer transforms the output hidden feature representations from the last transformer encoder layer into the change temporal patterns category labels.

There are several hyperparameters in the SAR-TST model, such as the number of the transformer encoder blocks N_{encoder}, the number of attention heads N_{head}, the model dimension $\text{Dim}_{\text{model}}$, and the dimension of the feedforward part of the encoder blocks Dim_{ff}. As shown in Table II, those values, which proved best in this study, were selected as defaults.

TABLE II
HYPERPARAMETERS SETTING IN THE PROPOSED SAR-TST MODEL

Parameter	Value
$N_{encoder}$	3
N_{head}	16
Dim_{model}	128
Dim_{ff}	256

As can be seen from Fig. 5, the classification model is first pretrained using unlabeled synthetic data to initialize the parameter weights. This self-supervised pretraining method is conducted with an autoregressive task of denoising the input samples. Specifically, 15% (empirical value) of the timesteps in a time series are randomly set to be 0, and the model is forced to predict these masked values according to contextual information. The model is therefore learning to model temporal dependencies between the timesteps of the time series.

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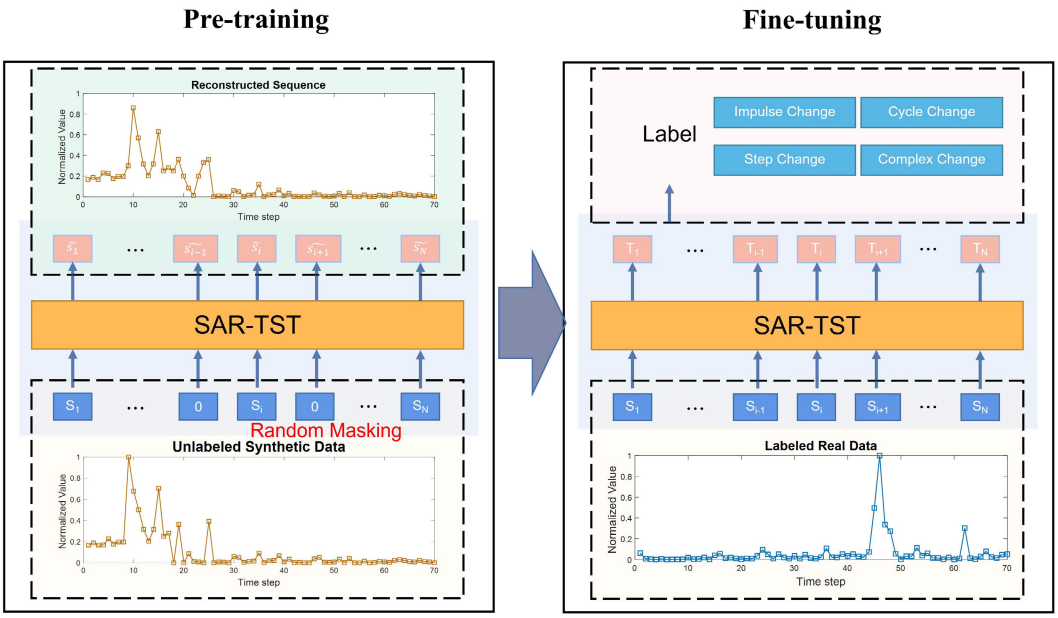


Fig. 3. Overall framework of the proposed classification model. Left: unsupervised pretraining model. It can be seen as an autoregressive task of denoising the input sequences. In this stage, all the input samples are from the synthetic data. Right: fine-tuning model. It can be seen as a generic TSC task. In this stage, the input samples are from the real-world data.

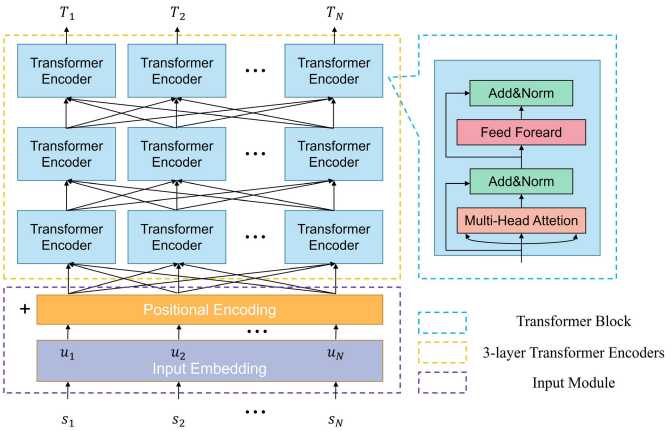


Fig. 4. Generic network structure of the proposed SAR-TST architecture. The normalized input time series $\mathbf{s}_{1:N}$ is linearly projected to a feature vector $\mathbf{u}_{1:N}$ and is then fed to the self-attention modules after adding the learnable position encoding.

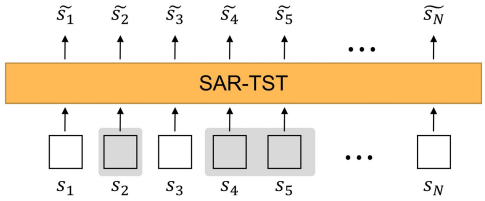


Fig. 5. Schematic of the proposed pretraining scheme. A proportion 15% of each input sequence is masked independently, and the model tries to predict full, uncorrupted input sequence. Noted that only the predictions on the masked values are considered in formula (19).

The mean-square-error (mse) loss between real values and predictions is used in each pretraining sample

$$\mathcal{L}_{\text{mse}} = \frac{1}{|M|} \sum_{s_t \in M} \|s_t - \tilde{s}_t\|^2 \tag{19}$$

where M denotes the set of masked values. In the fine-tuning step, the model parameters will be updated by feeding the constrained real-world samples.

IV. Experiment

A. Case Study and Data Source

The proposed method was applied to the urban area in the northwest of Hong Kong, China, between 22°26′N–22°29′N and 113°58′E–114°21′E, as shown in Fig. 6. The case area mainly belongs to Yuen Long, New Territories, a brand-new development zone adjacent to Shenzhen. It is surrounded on three sides by mountains, while the land is flat in the central region. The vast areas of seasonal fishponds are widely distributed in the north area, which is the main aquaculture base. With the rapid development of industrialization in recent years, the villages, old low-rise and newly built high-rise residential buildings are all mixed up in the middle area [60].

Hong Kong's climate is subtropical, characterized by long, hot, and rainy summers, and cool, humid winters. SAR remote sensing data played a critical role in LCLU monitoring in rainy and cloudy conditions, which are more suitable for Hong Kong's environments. The SAR ITS data used in this study are from the CSK satellite sensor, which had a spatial resolution of 3 × 3 m in the stripmap mode and a spatial coverage of 1600 km^2. The dataset that consisted of 70 HH Level-1A descending CSK images was acquired from 2013 to 2020 with a time resolution of nearly one month. The basic parameter information of the dataset is shown in Table III. From this long time series, not only the significant changes can be observed, but also the different change patterns can be identified, e.g., seasonal changes in fishponds and step changes in residential areas.

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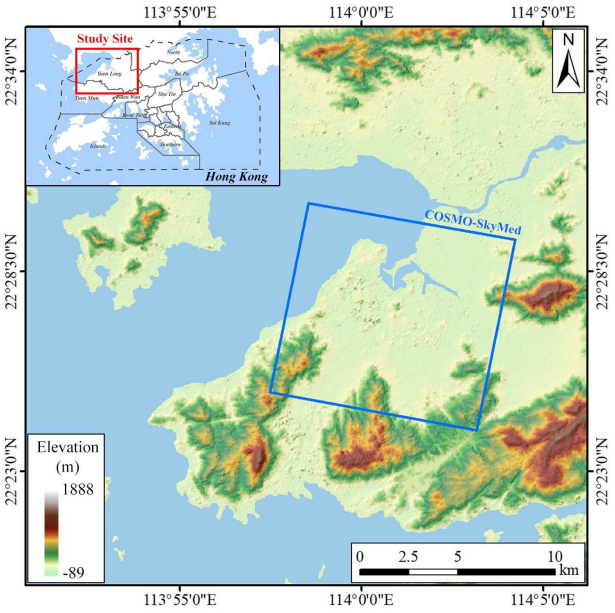


Fig. 6. Geographic location of the study area.

TABLE III
BASIC INFORMATION OF THE CSK DATASET

Specifications	Cosmo-Skymed Dataset
Operator	Agenzia Spaziale Italiana (ASI), Ministero della Difesa (MDD)
Satellite	Cosmo-Skymed constellation (4 Satellites)
Launched date	June 08, 2007
Minimum repeat cycle	4 days
Band	X-band
Imaging mode	Stripmap
Data product	SLC Level 1A-SCSB
Resolution	3m×3m (range×azimuth)
Polarization	HH
Image number	70 (descending orbit)
Acquisition dates	2013/10/04 ∼ 2020/07/14
Size of study area	5098×4622

B. Change Detection Results and Analysis

In the following, we will give a quantitative analysis for the proposed SAR-PELT algorithm and then try to analyze the change regulations in the study area.

1) Performance Evaluation of the SAR-PELT Algorithm: To evaluate the performance of the proposed SAR-PELT algorithm, the benchmark SAR ITS CPD dataset was constructed, which consisted of three typical change patterns: impulse, step, and cycle. Each pattern contains 80 samples, and the total of 240 labels was created manually by referring to the suggestions from expert assessment. It is worth noticing that every sample contains at least one change point.

The OA was used as the main evaluation metric. Considering that the change points have a certain temporal vagueness, the tolerance interval was adopted to reduce the absolute error, which is set to be 3 in this study. Besides, we defined the average Hausdorff distance ($\overline{\text{Hd}}$) to evaluate the average accuracy [61]. Formally, it is equal to the average temporal distance between the observed value and the real value nearest to the observed value

$$\overline{\text{Hd}}(\hat{\mathcal{T}}, \mathcal{T}^*) = \overline{\sum \min \left|\hat{\tau} - \tau^*\right|} \tag{20}$$

where $\hat{\mathcal{T}}$ represents the real change points indexes and $\mathcal{T}^*$ represents the detected indexes. The lower the $\overline{\text{Hd}}$ is, the higher the accuracy is.

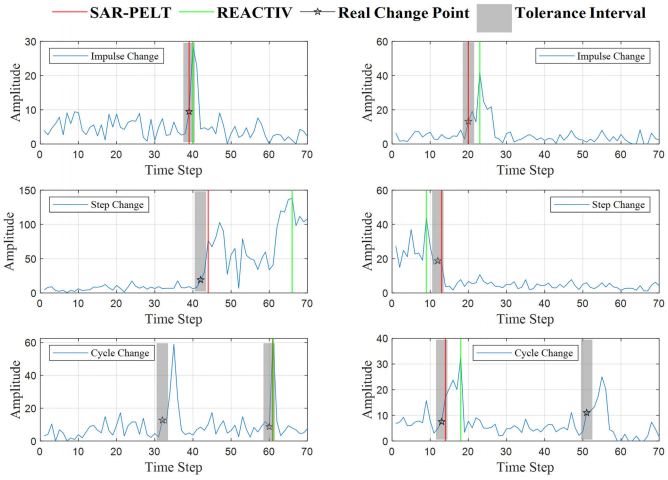


Fig. 7. Comparison of the different CPD methods on real-world samples. We randomly select two samples for each class (impulse, step, and cycle).

TABLE IV
RESULTS OF QUANTITATIVE EVALUATION ON CPD

Methods	OA (Impulse)	OA (Step)	OA (Cycle)	OA (All)	$\overline{Hd}$
REACTIV	0.8750	0.0750	0.6875	0.5458	4.7458
SAR-PELT	**0.9250**	**0.8625**	**0.8000**	**0.8625**	**2.4083**

The REACTIV algorithm was also introduced for comparison. As a popular SAR ITS change detection method, it has been applied to urban area change monitoring. Fig. 7 shows the detection results in six randomly selected samples. Intuitively, the change points detected by SAR-PELT are closer to the real change points compared with REACTIV. The results of the quantitative evaluation are shown in Table IV. The OA of SAR-PELT reached, on average, 0.86, and the accuracy of each type was over 0.80, which are much higher than REACTIV. Moreover, the performance of REACTIV was deplorable, especially in step change patterns. Indeed, the strategy of REACTIV to determine the change points is to find the maximum amplitude point over the time series, which is very friendly for short impulse change. However, it is permanently stable for the step pattern after the change occurs, the maximum amplitude point is usually far from the tolerance interval, which results in considerable error. In addition, compared with REACTIV, SAR-PELT has a lower $\overline{\text{Hd}}$, which means that the prediction of SAR-PELT is much closer to the real value.

2) Change Regularity Analysis: Fig. 8 shows a general view of the change dynamics over the time in the observed area using the HSV model. Different hues represent the changes that occur at different times and saturation represents the change frequency during the observation period. The change temporal information and the change frequency information

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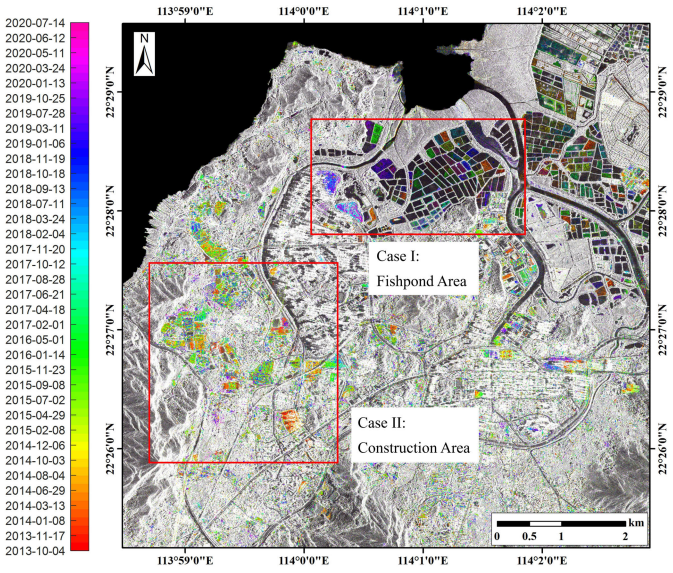


Fig. 8. General view of the change dynamics in the study area. Two main change areas are marked by the red rectangle. Case I is the fishpond area and Case II is the construction area. Please note that the number of time points is 70, and here, we list a half on the left time axis.

are comprehensive and vividly shown in the HSV model. Case I in Fig. 8 is a study fishpond area, and case II is a typical construction area. Fig. 9 shows the enlarged maps of both study areas. From Figs. 8 and 9, due to the diverse topography in the study area, the land cover types are considerably varied. The temporal–spatial distributions are inevitable showing significant regional differences.

a) Change spatial distribution characteristics: The maps of change frequency in Fig. 10 show the variation degree in the study area, and each pixel in the maps has a CV value that reflects the level of change frequency from 0 to 1. It is easy to see that the change spatial distribution is closely bound up with the land cover types. In detail, the high-frequency changes are mainly occurring in the densely populated construction areas located in the midwest and some fishpond areas located in the north (CV value ≥ 0.7), which are closely correlated with human production and life activities. With the rapid development of local industrialization in Yuen Long, population growth and urbanization have been accelerated in this period (2013–2020), and a huge amount of construction has been newly built along the highway in the midwestern plain [62]. For the fishponds, the existence of the dry season and the periodic operation of draining off water lead to frequent land cover changes. The other areas, especially the southern and eastern mountain areas, preserved a lower frequency and slight change trend (CV value ≤ 0.4). Due to the fruitful forest conservation policy of the Hong Kong government, large tracts of forest resources are protected from destruction by human activities, which maintains a relatively stable state without any land cover changes in a long time.

b) Change temporal distribution characteristics: Different land cover areas also present different change temporal distribution characteristics. Fig. 11 shows the annual statistic histogram in Cases I and II. As can be seen from Fig. 11(a), the overall distribution is more even in the fishpond area. From a seasonal perspective, the number of change points in autumn

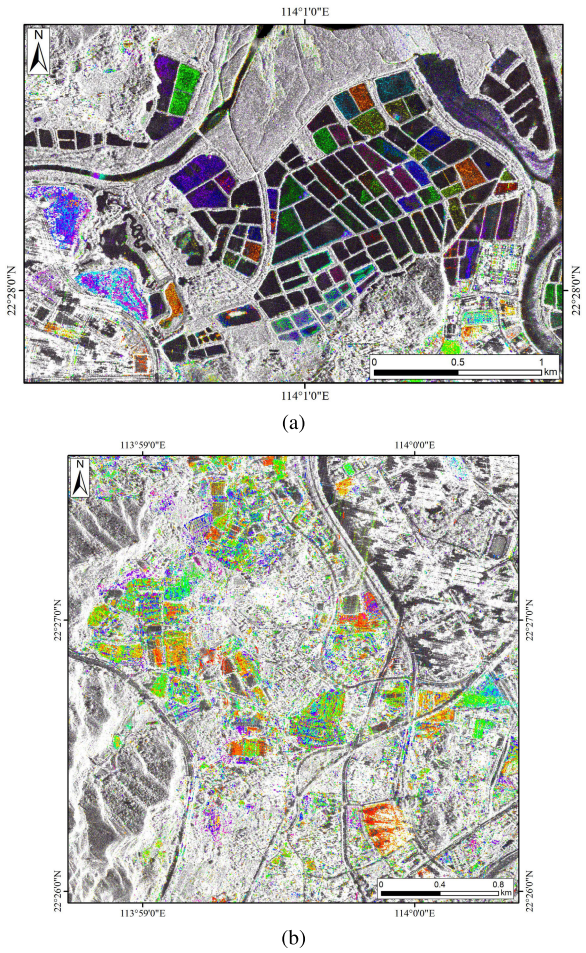


Fig. 9. Locally enlarged drawings of the change dynamics in the areas of Cases I and II. (a) Case I: fishpond area. (b) Case II: construction area.

is noticeably higher than in other seasons. The fishermen usually conduct the fishing operations from July to September and clear the waste water after that. For the construction area, the interannual fluctuation of the change points varied widely. It shows a gradual downward trend from 2013 to 2020. In particular, the highest number of change points occurred in 2013 and 2014. However, in the years that followed, the urban land use in Yuen Long was nearly saturated, and the number of change points was significantly reduced. From the analysis of the temporal–spatial distributions, human factors made a more dramatic difference than the natural factors for land cover changes.

C. Change Pattern Analysis

1) Experiment Settings: In this study, the total of 60 000 pretraining synthetic samples was constructed, which was used for learning the generic time series representation and initializing the parameter weights. For real-world labeled SAR ITS samples, we introduced the data augmentation with flipping and noise addition; thus, a total of 13 980 samples were used for fine-tuning the pretrained model. To evaluate the effectiveness of the proposed network model, we compared it with some state-of-the-art models, which are widely implemented in TSC tasks. For RNN-based architectures, we selected classical LSTM [63] and gate recurrent unit (GRU) [64]. For CNN-based architectures, we selected ResNet [65] and

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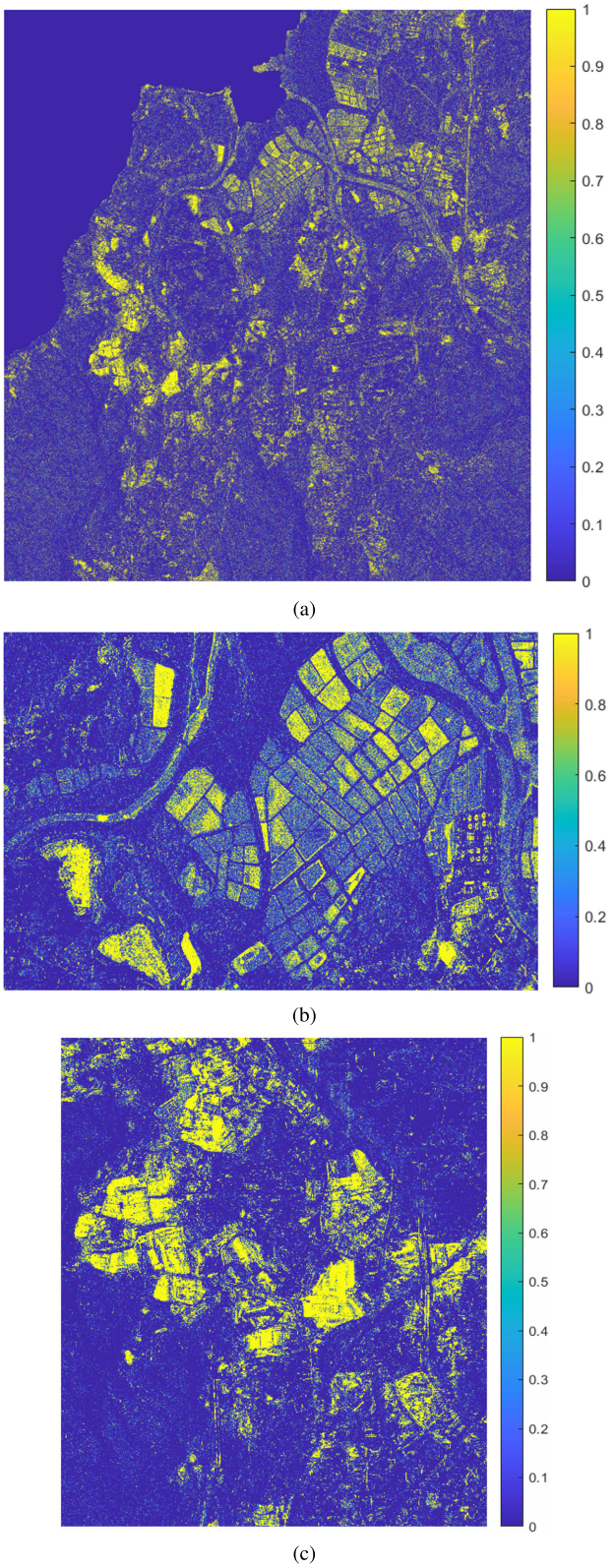


Fig. 10. Change frequency maps based on CV statistic. (a) General view of the study area. (b) Case I: fishpond area. (c) Case II: construction area.



Fig. 11. Annual statistic histogram of the change points on the area of Cases I and II. (a) Case I: fishpond area. (b) Case II: construction area.

InceptionTime [66]. All of them have shown good properties on TSC tasks.

1) *LSTM:* This LSTM model is formed by three stacked Bi-LSTM layers that can encode SAR ITS to increasingly higher level representations. The hidden states from both forward and backward directions are concatenated and used as the input of the next layer.
2) *GRU:* This GRU model is formed by three stacked Bi-GRU layers. Similar to Bi-LSTM, it can learn important temporal information from the current input bidirectionally.
3) *ResNet:* The ResNet-based TSC architecture proposed in [65] is consisting of three separate streams stack, each of which has three residual blocks with increasing convolutional kernel sizes $\{3, 5, 7\}$. Therefore, this model is good at extracting different temporal scale features.
4) *InceptionTime:* The proposed InceptionTime model in [66] is an ensemble of five Inception networks rather than a single network. This ensemble architecture is well suited to more complex time series and is available to control and reduce the classification error.

The performances of different classification models are compared using the following three main metrics, which are derived from the confusion matrix: OA, F1 score, and Kappa coefficient. In our experiments, all the models were pretrained for 200 epochs, and the batch size is set to be 256. We utilized the Adam optimizer with a learning rate of $1e^{-3}$ for SAR-TST and $2e^{-3}$ for other models. Also, we fine-tuned the model for 50 epochs in a total-supervised manner with a learning rate of $2e^{-3}$. Every model repeated the test ten times and averaged the results. All the models are implemented on the Python Pytorch 1.11 platform with an Intel Core i7-11800H CPU and a NVIDIA GeForce RTX 3060 Laptop GPU.

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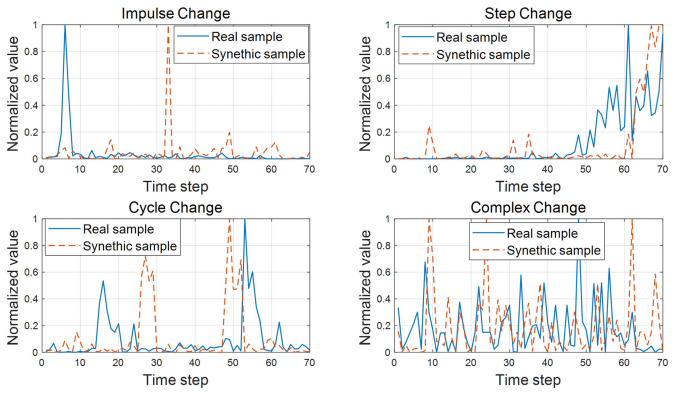


Fig. 12. Comparison of the randomly selected real-world samples and synthetic samples on four change patterns.

2) Quality Evaluation of Synthetic Data: It is necessary to evaluate the effectiveness of the synthetic data before application. The easiest approach to qualitative evaluation is to simply plot the synthetic time series and compare them to the real time series. Fig. 12 shows the comparison of the randomly selected real samples and synthetic samples in terms of their visual appearance. Intuitively, there is no significant difference in the varying trend of the time series with the same change pattern, though some parameters, such as variation extent and duration, are changed. The ideal synthetic data can further broaden the distribution information of original samples and enhance their diversity. However, comparing the waveform directly is difficult to see the smaller differences of the distribution between synthetic and real data. To tackle this issue, we introduce the t-distributed stochastic neighbor embedding (t-SNE) method to visualize the distribution characteristics [67]. Fig. 13 shows the scatter plots of the synthetic and real data on four change patterns using t-SNE for better visualization. In Fig. 13(a)–(c), it is obvious that the distribution of the real data is constrained within relatively small bounds, while the synthetic data heavily broaden the distribution. The scatters in Fig. 13(d) are chaotic and random, as the complex samples are irregular. Fig. 13(e) shows the scatters of the overall samples, and we can see that the synthetic data are not well overlapped with the real data, which demonstrates that there are still some feature differences between them. In other words, synthetic data can be seen as simplification in an ideal state, and it cannot fit the distribution of the real data completely. Considering that the purpose of using synthetic data is to improve the performance of the classification, rather than generating samples as close to the real data as possible, the generation samples are able to satisfy the requirements of downstream classification tasks.

To quantitatively evaluate the effectiveness of synthetic data, we conducted two supervised tasks, one called "train on synthetic, test on real (TSTR)" and the other called "train on real, test on synthetic (TRTS)" [68]. For TSTR, we use synthetic data to train a supervised classification model and then test it on the real data. This task can evaluate the ability of synthetic data in real-world applications. For TRTS, on the contrary, the real data are used for training the model and then test it on synthetic data. This task is used to evaluate how realistic the synthetic data are and how much the distribution differs between them. Table V shows the results of the quantitative

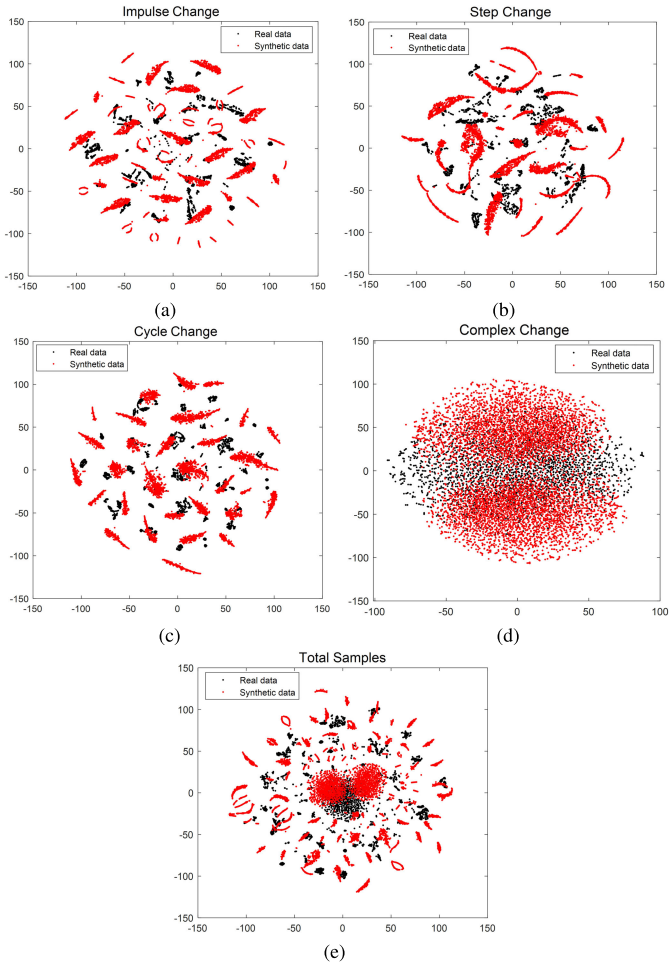


Fig. 13. Scatter plots using t-SNE visualization on (a) impulse change samples, (b) step change samples, (c) cycle change samples, (d) complex change samples, and (e) total samples using t-SNE method.

TABLE V
RESULTS OF QUANTITATIVE EVALUATION FOR SYNTHETIC DATA

Methods	OA	F1	Kappa
TRTR	0.982	0.984	0.967
TRTS	0.533	0.494	0.378
TSTR	0.732	0.747	0.646

evaluation using the SAR-TST model, and we use "train on real, test on real (TRTR)" as the benchmark. The model achieved an OA of 0.982 in TRTR, while the performance is significantly reduced as the OA in TRTS is only 0.533. The reason is due to the insufficiency of the real samples, as the learned features are unable to identify the change patterns in synthetic data. For TSTR, the OA is 0.732, which is much higher than TRTS. This means that the model can learn more general temporal features from synthetic data and thus improve the ability of the classification, which further verifies the necessity of introducing the synthetic data. In a word, synthetic data are an important complement to enhance data diversity and further improve the generalization of the model.

3) Performance Evaluation of the Training Scheme: To evaluate the performance of the proposed SAR-TST model, we conducted comparison experiments using real-world samples in different proportions, which were gradually promoted

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TABLE VI
QUANTITATIVE TEST RESULTS OF DIFFERENT PRETRAINED NETWORKS

Proportion	Methods	OA	F1	Kappa
10%	LSTM	0.828	0.829	0.759
	GRU	0.895	0.893	0.851
	ResNet	0.935	0.935	0.913
	InceptionTime	0.958	0.958	0.944
	SAR-TST	**0.976**	**0.977**	**0.965**
50%	LSTM	0.948	0.948	0.926
	GRU	0.953	0.954	0.938
	ResNet	0.993	0.992	0.990
	InceptionTime	0.994	0.995	0.992
	SAR-TST	**0.997**	**0.997**	**0.996**
100%	LSTM	0.977	0.979	0.959
	GRU	0.974	0.975	0.966
	ResNet	0.997	0.997	0.997
	InceptionTime	**0.999**	**0.999**	**0.999**
	SAR-TST	**0.999**	**0.999**	**0.999**

from 10% to 100%, as they were all pretrained by synthetic data before. Fig. 14 shows the change curves of three evaluation metrics for different proportions of the input samples. Table VI shows the quantitative evaluation results of different models when the proportion is set to be 10%, 50%, and 100%. The best value of each metric is shown in bold. Intuitively, as the proportion increases, three types of curves for all the models increase rapidly, and then, the growth rate slows down and converges to a constant value gradually. In detail, SAR-TST achieved the best performance as the proportion was less than 50%, which means that the learned global generic features in the pretraining step effectively promoted the downstream classification task in the conditions of insufficient real data. In particular, when the proportion is 10%, the OA of SAR-TST is over 0.97, which is much higher than the comparison models. CNN-based InceptionTime and ResNet take second place, while RNN-based LSTM and GRU have yet shown satisfying performance. The reasons are mainly due to the special integrated network structures of InceptionTime and ResNet, which could acquire more temporal features and reduce the classification error. Another important finding is that, when the proportion is up to 100%, all the DL-based methods achieve very high classification accuracy, especially the SAR-TST and CNN-based methods, whose OA are all greater than 0.99. The number of input real samples fully meets the requirements for fine-tuning at this time. Moreover, when the proportion is 60%, SAR-TST has already reached 0.99 of OA, which means that it achieves faster convergence with a smaller real data size. Considering that we focused on four main change patterns, the results are convincing in the conditions of utilizing adequate pretraining samples. SAR-TST is still the first choice, especially under realistic conditions lacking real-world samples.

In addition, the proportion of the synthetic data used in the experiments is another important parameter. On the one hand, too many synthetic data could lead to the redundant feature information and increase the training cost. On the other hand, insufficient synthetic data could reduce the performance of

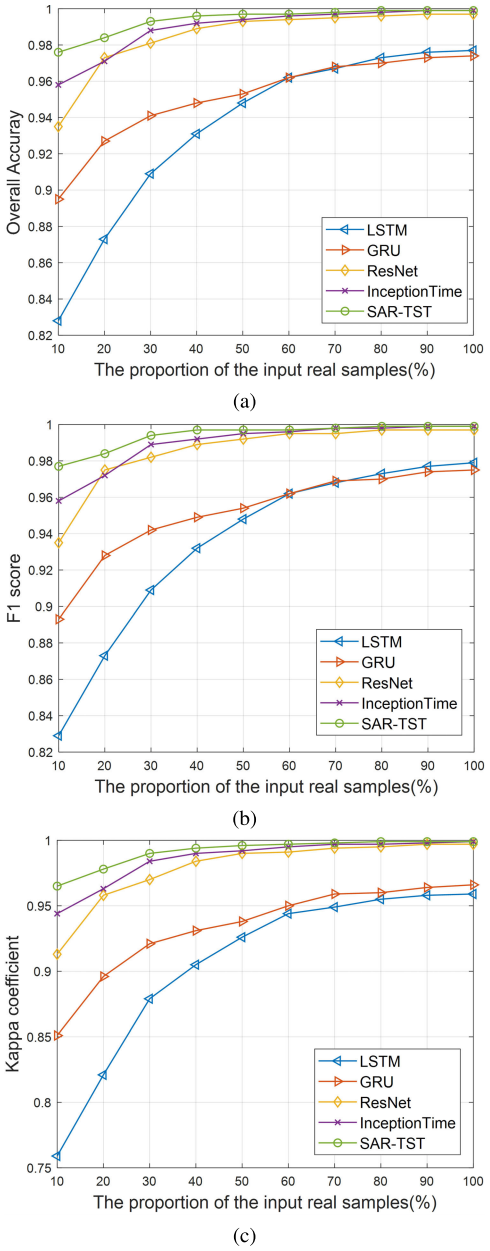


Fig. 14. Performance of the proposed SAR-TST and comparison models using different proportions of the input real samples. Evaluation metrics: (a) OA, (b) F1 score, and (c) Kappa coefficient.

the pretrained model. To obtain the best proportion of the synthetic to real data, comparison experiments were performed using the SAR-TST model, fixing the real-world samples and adjusting the proportion of the synthetic samples with respect to total number of real samples, from 1% (600 samples) to 100% (60 000 samples). Fig. 15 shows the change curves of the OA for different proportions of the input synthetic samples. It can be observed that, when the proportion of the real samples is less than 30% of their total, the OA significantly increases as the proportion of the synthetic samples raises from 10% to 50%. Instead, when the proportion of the synthetic samples is more than 50% of their total, the OA no longer grows, which indicates that the model cannot learn more effective change features no matter how large the size of the synthetic data is. In other words, the change features extracted

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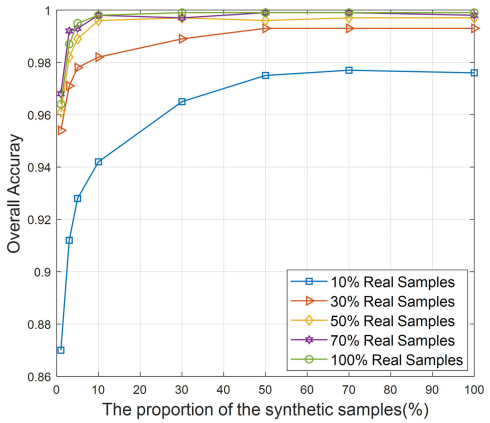


Fig. 15. Line chart of the synthetic samples proportion and OA with the fixed sizes of the real samples. The test proportions of the synthetic samples include 1%, 3%, 5%, 10%, 30%, 50%, 70%, and 100%.

TABLE VII
COMPUTATIONAL EFFICIENCY OF DIFFERENT PRETRAINED NETWORK

Networks	Training Time(200 epoches)	samples/Second
LSTM	3265s	2940
GRU	2796s	3433
ResNet	5904s	1626
InceptionTime	9640s	995
SAR-TST	13166s	730

from the added synthetic samples are redundant. Besides, when the proportion of the real samples is more than 50%, the results revealed that the OA improves more than 3% if the proportion of the synthetic samples is less than 10% and there are no longer any changes for more than 10%. The rationale is that the change features extracted from the real samples are more effective to TSC compared with the generic features extracted from the synthetic samples. In this study, we used 50% synthetic data for the pretraining task. Considering that real data are usually too few, using a large set of synthetic data is still the best choice.

The total elapsed time for different networks in the pretraining stage (using 100% synthetic data) is shown in Table VII, and we compared their training speeds on a single GPU. It can be observed that obviously, GRU had the fastest computation speed, followed closely by LSTM, and InceptionTime and SAR-TST are the two slowest ones. The deeper network architecture and the huge number of parameters of SAR-TST lead to the slowest training speed. However, they allow for more parallelization, which can lead to significant gains in computation efficiency. Indeed, SAR-TST can conduct massive parallel computations on multiple GPUs, which is more adapted to process large-scale SAR ITS.

4) Classification Results on the Study Area: Fig. 16 shows different change pattern cases in the study areas, where the optical images are acquired from historical satellite imagery in Google Earth and SAR images are obtained from CSK data. As can be seen from Fig. 16(a), the fishpond area inside the red box had changed markedly as an impulse pattern from 2013-10 to 2019-02. The residential area in Fig. 16(b) began construction on 2013-11 and fully completed on 2014-11. This process of starting from scratch is a typical step change.

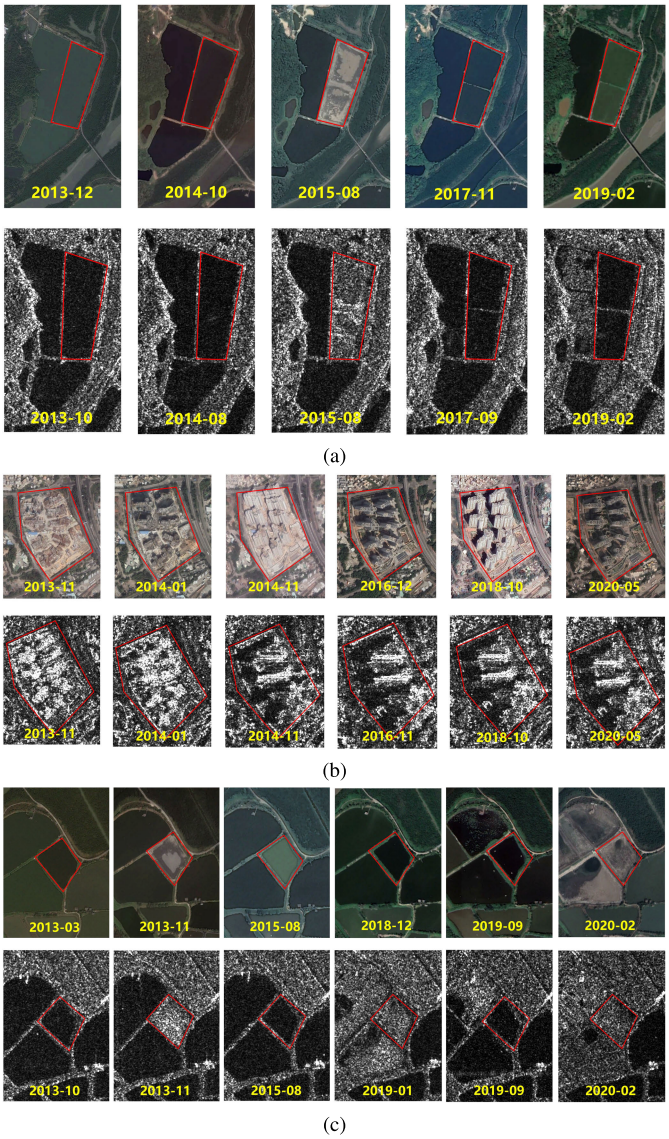


Fig. 16. Optical and SAR images of different change pattern cases from the study area. (a) Region of impulse change. (b) Region of step change. (c) Region of cycle change.

In Fig. 16(c), there are three significant land cover changes in 2013, 2019, and 2020 highlighted in a red box, and thus, it is a typical cycle change.

Fig. 17 shows the result of the change pattern classification on the study area using the HSV model. Here, we used 100% labeled data to fine-tune the SAR-TST model. Fig. 18 shows the local enlarged drawings of the fishpond area and construction area. Generally speaking, different land cover areas present different characteristics for change pattern distribution, as there are significant visual differences between the fishpond area and the construction area. In detail, there are more impulse and cycle changes in the fishpond area, which are closely related to human periodic activities. On the contrary, there are more step changes in the construction area. The development of the construction areas experienced one to grow out of nothing, which is a typical permanent step change. Some regions maintain a relatively stable state, such as forest areas. The classification results, to some extent, prove that human activities and technological development accelerate

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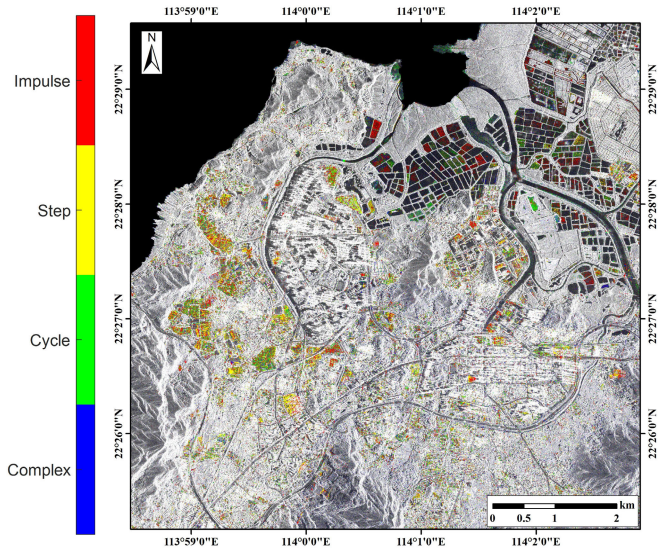


Fig. 17. General view of the change pattern classification result on the study area.

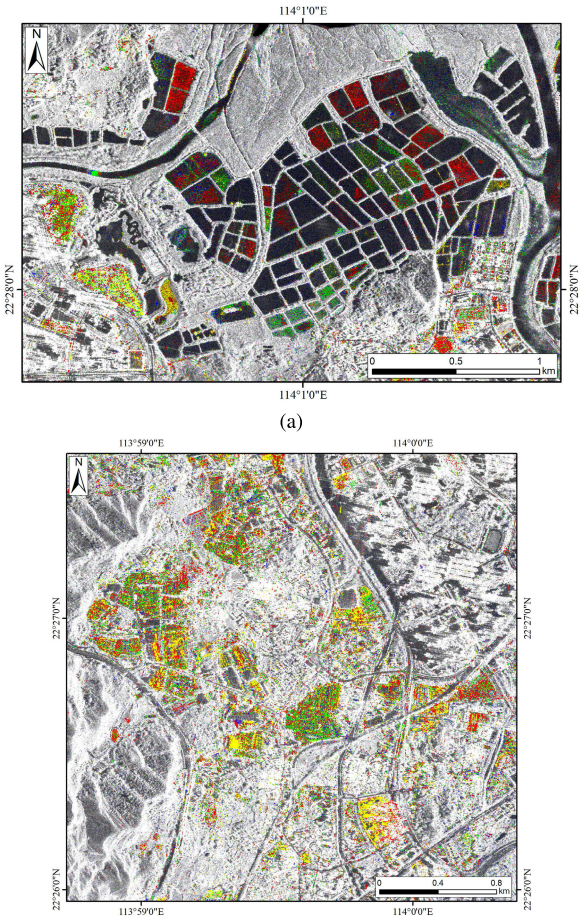


Fig. 18. Locally enlarged drawings of the change pattern classification in the areas of Cases I and II. (a) Case I: fishpond area. (b) Case II: construction area.

LCLU change, while the influence of nature factors (excluding sudden nature disasters) is not obvious. The nature of land cover changes will be a long-term evolution process, perhaps on a decade scale or more.

V. Conclusion

In this article, we proposed a novel unified framework named long time series SAR image change detection and change pattern analysis (SAR-TSCC) to monitor LCLU change dynamics in the large-scale urban area. The proposed mapping scheme is based on the HSV model. We first developed the SAR-PELT algorithm to extract the change time information quickly. Meanwhile, the SAR-TST model was used to classify the change in temporal patterns. The model parameters were initialized by pretraining on abundant synthetic samples in an unsupervised manner. Then, the model was fine-tuned using the limited labeled real-world data, which can help the model better adapt to the study area. The proposed framework was used for monitoring a large urbanization area in the northwest of Hong Kong, China. Our approach achieved a high accuracy of 86% in detecting change times and 99% in classifying the change patterns. Land cover change information in the study area can be effectively identified, and it is of great guiding significance for urban managers to make socioeconomic policies and environmental protection plans.

In this study, we validated the effectiveness of the proposed framework on single spatial resolution data. To further expand the applications of the proposed framework, using multisource SAR data with higher temporal–spatial resolution to implement more accurate and fine-grained monitoring is the focus of our future research.

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